

IBAD ULLAH 25K-6544

CE 2-A PF LAB TASK 01

OBJECTIVES:

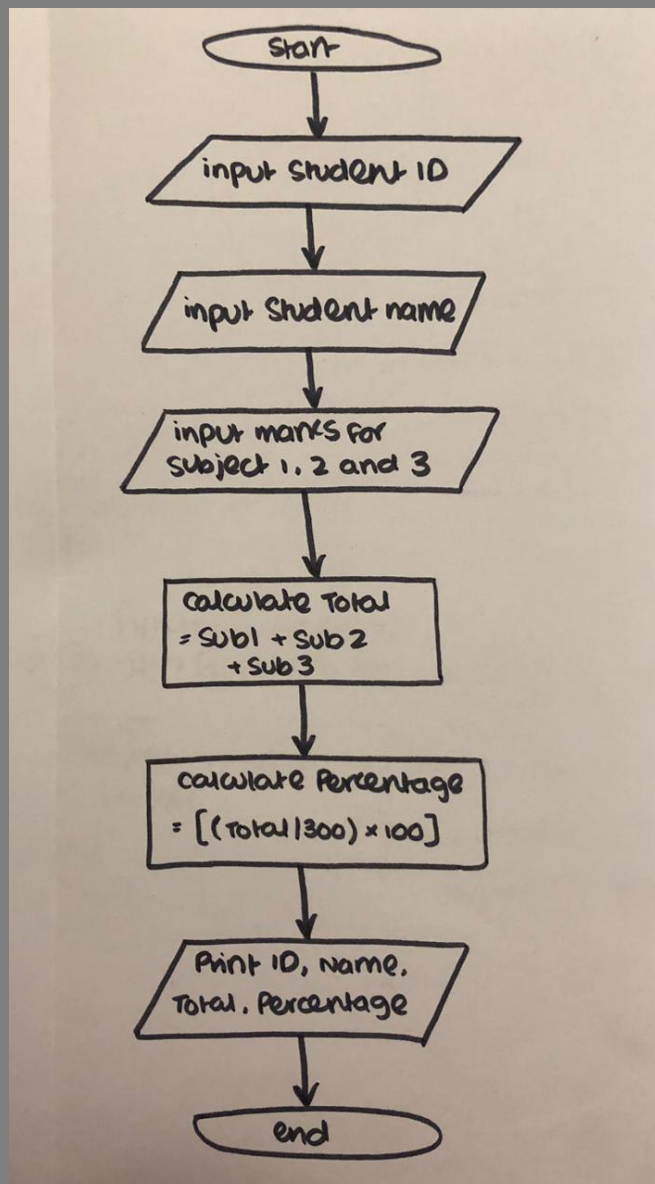
Learn the basics of programming and practice solving problems using real-life examples.

Learn how to read and draw flowcharts for programs.

Practice writing algorithms to show the steps and logic of a program.

Get hands-on experience with Scratch programming to apply these concepts.

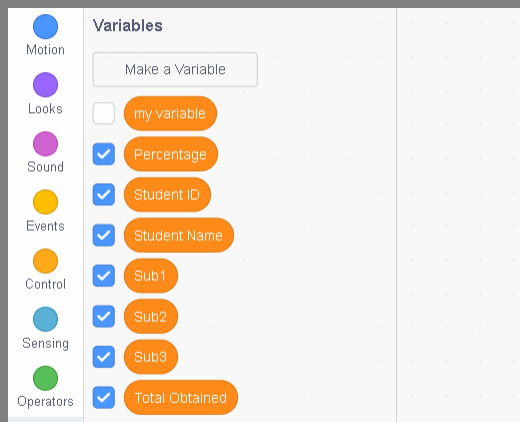
QUESTION 01



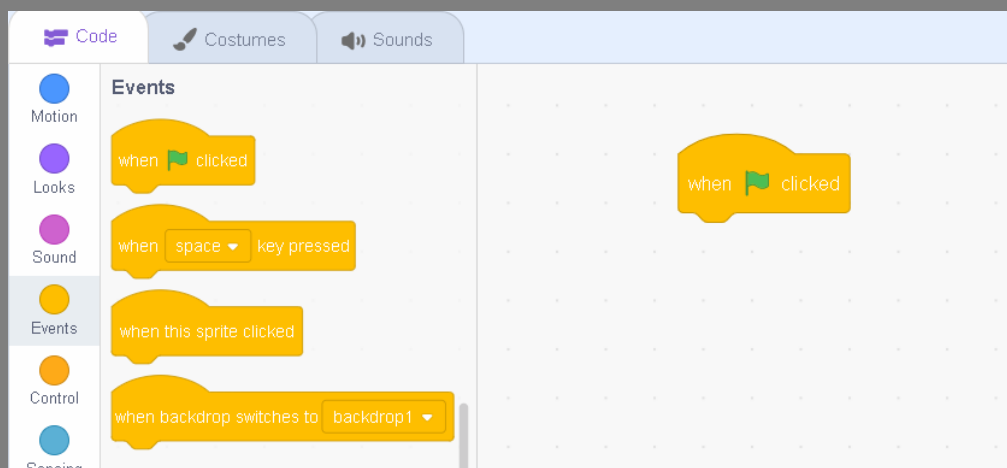
1. Start the program.

2. Declare Variables: Name, ID, Subject1, Subject2, Subject3, Total, and Percentage.
3. Prompt the user to input their ID and Name
4. Ask the user for marks in Subject 1, Subject 2, and Subject 3 one by one and store them.
5. Add the marks of all three subjects together and store the result in Total variable.
6. Divide the Total by 300 then multiply by 100 and store this result in the Percentage variable.
7. Display the Student's ID, Name, Total Marks, and Percentage.
8. End the program.

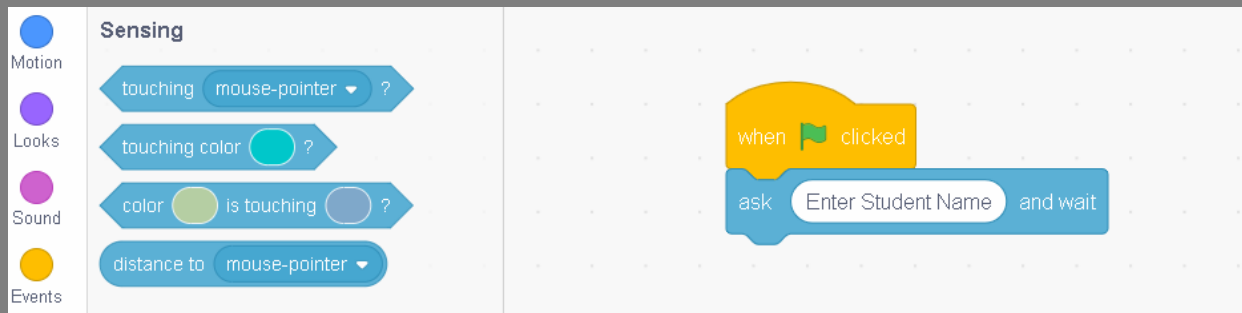
Creating variables to store data



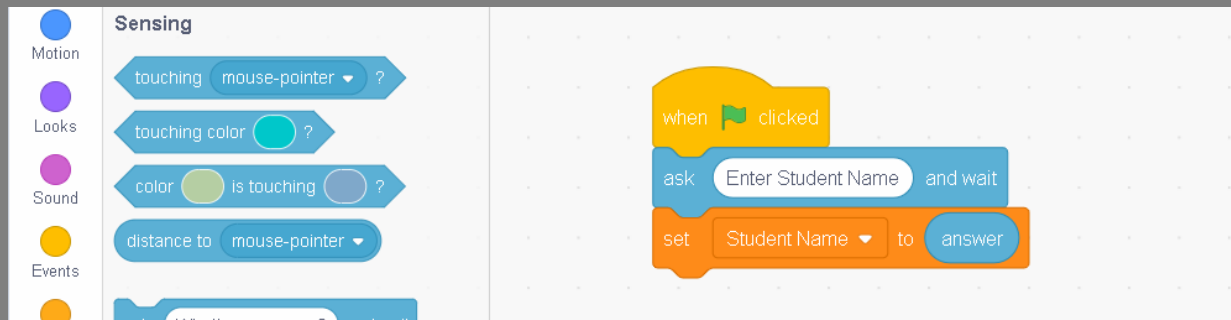
To start the program when the green flag is clicked



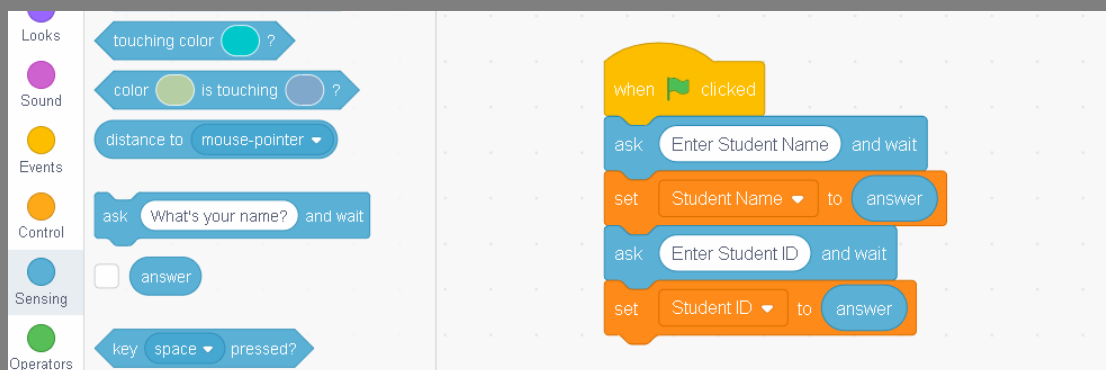
Prompts for name



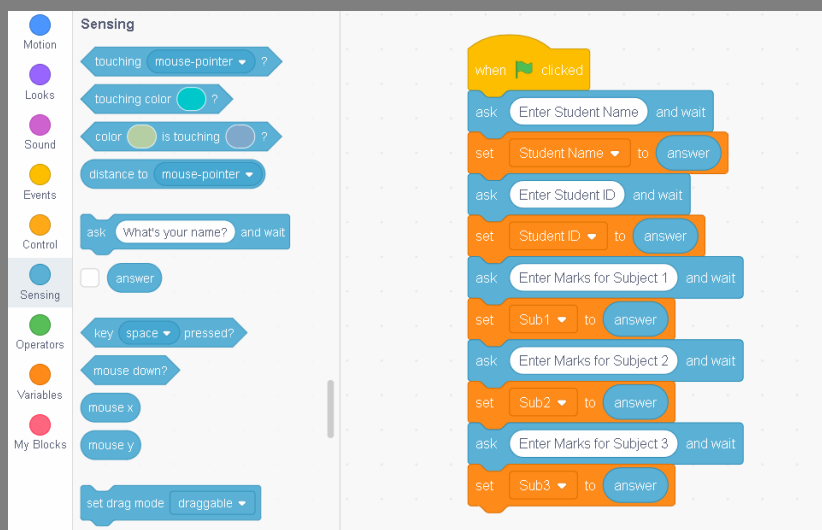
Saves the name



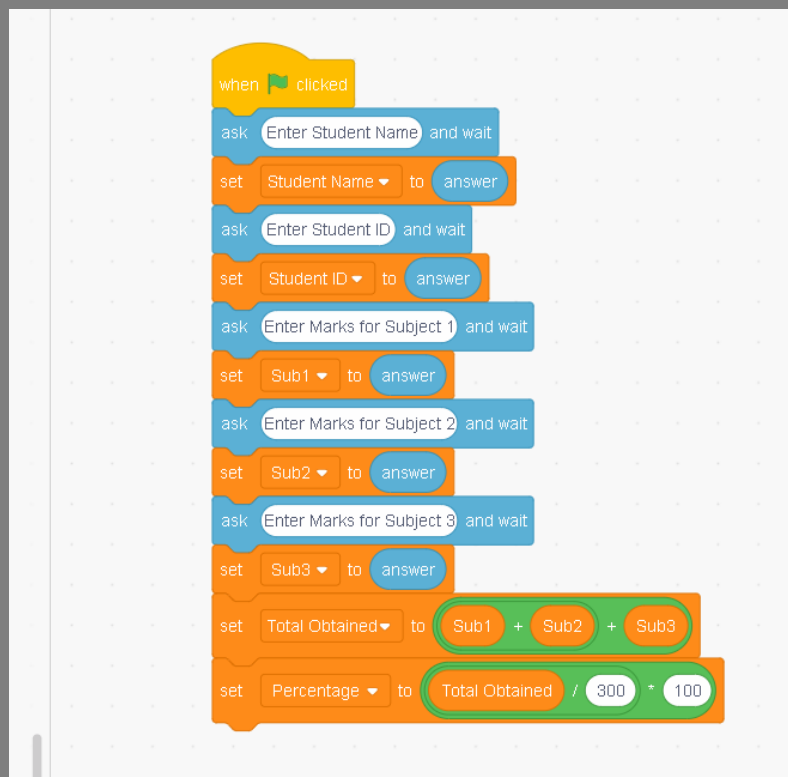
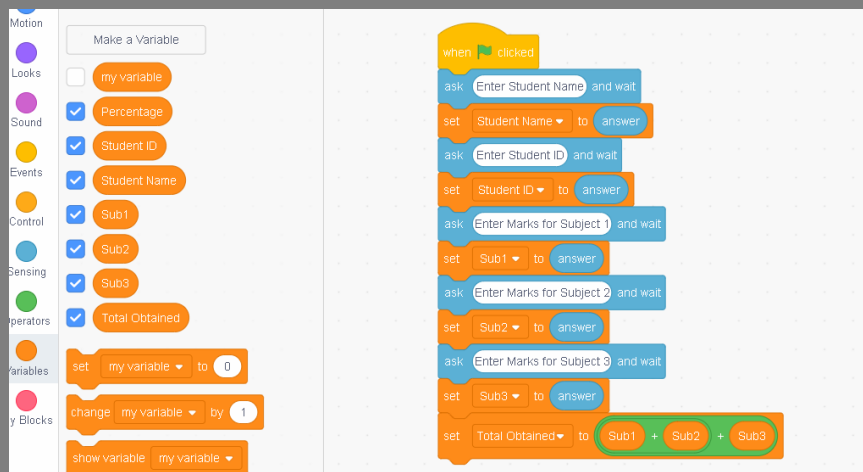
Prompts for ID and saves it



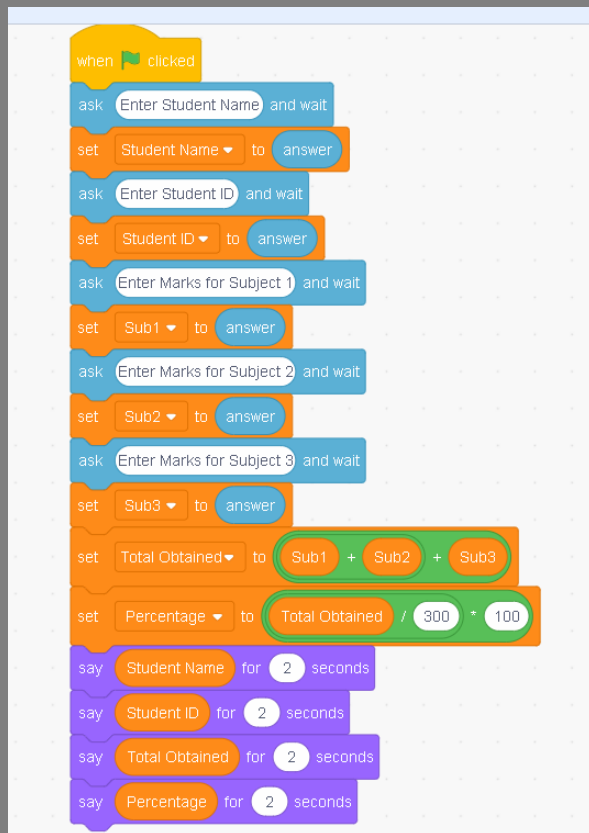
Prompts for marks for subj 1, 2, 3 and saves it



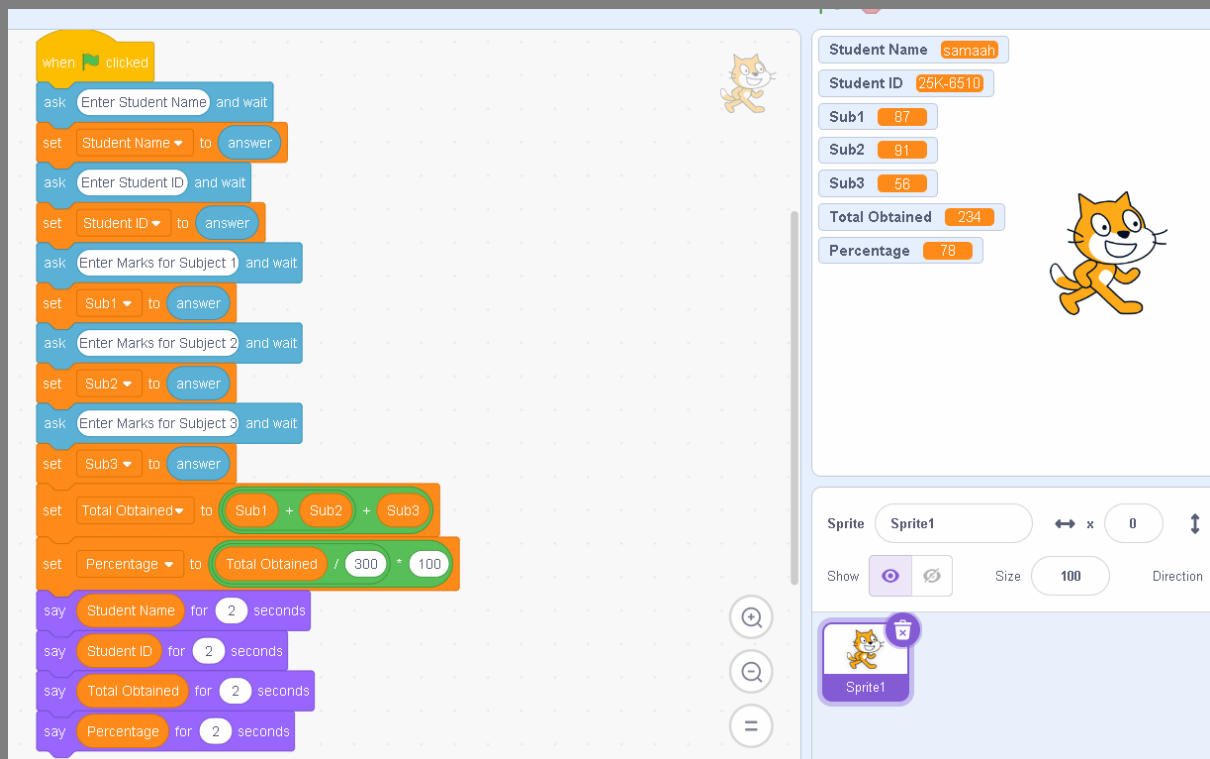
Building the formula block to calculate total and percentage



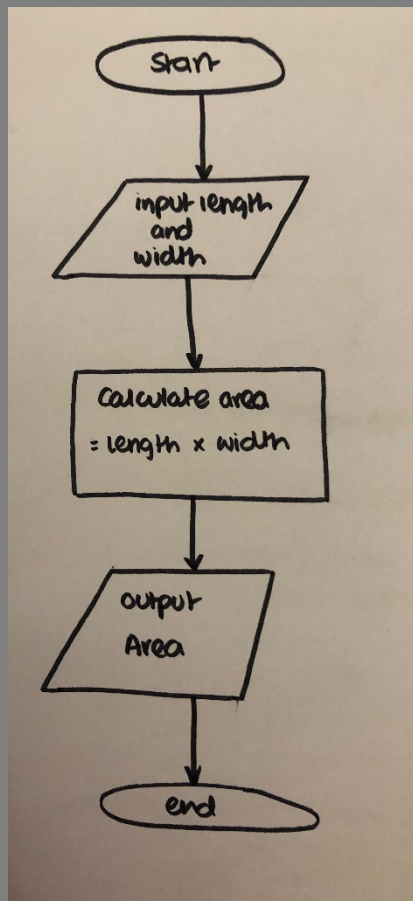
Character will speak student name, ID, total obtained and percentage



Output:

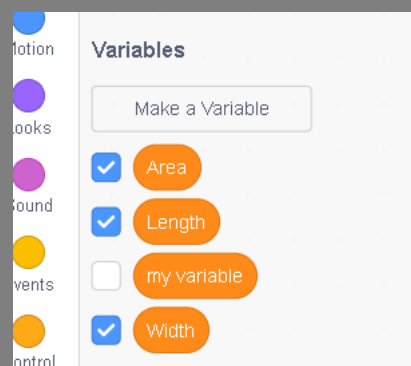


QUESTION 02



1. Start the program
2. Declare 3 variables: Length, Width, and Area.
3. Ask the user to enter the value of Width
4. Ask the user to enter the value of Height
5. Set Area = Width × Height
6. Display output Area
7. End the program

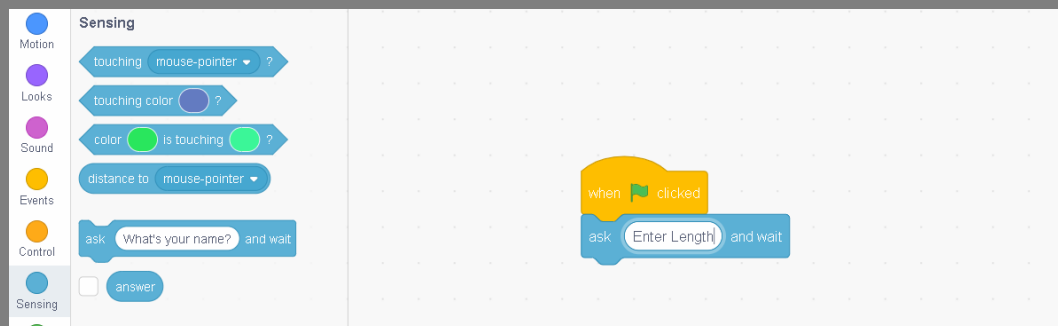
Creating variables that will store data



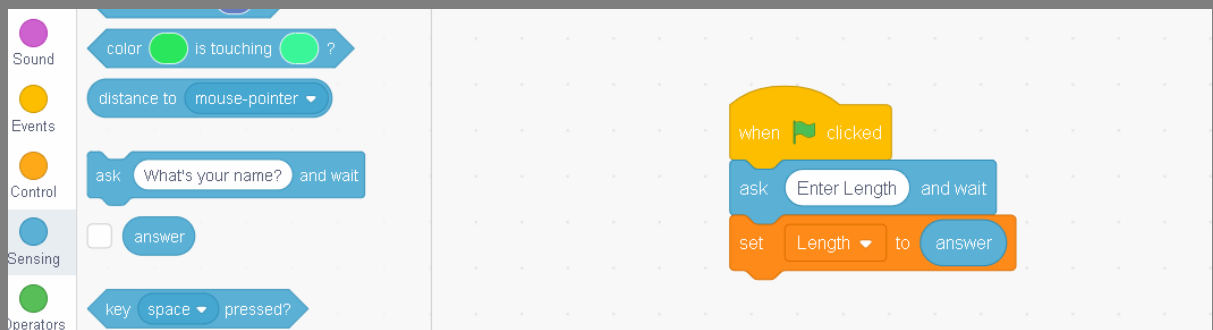
To start the program when you click the green flag



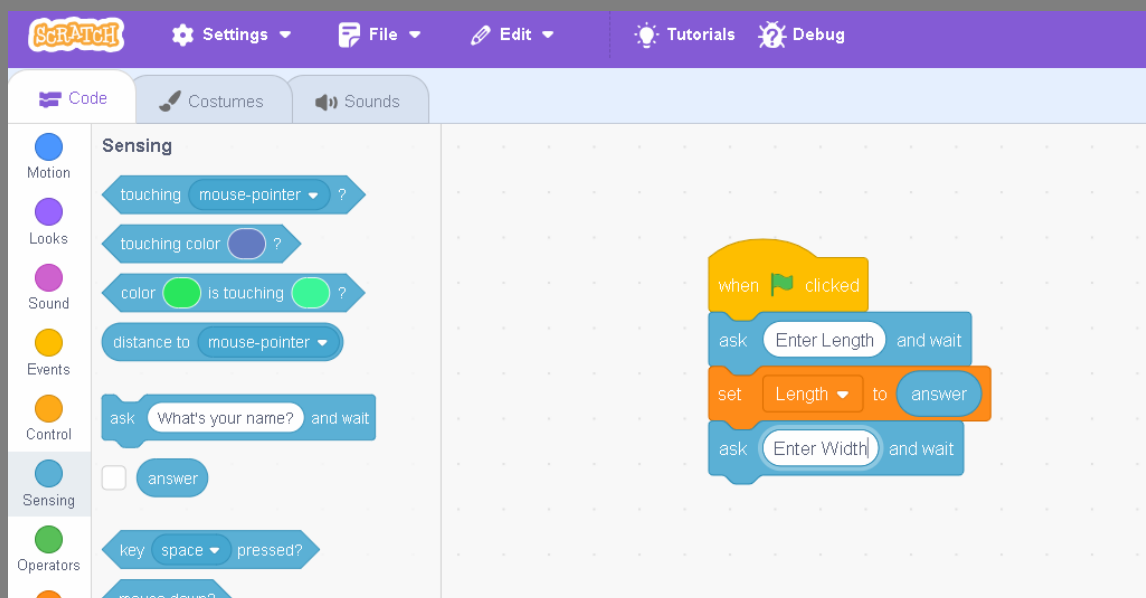
Prompts the user for Length



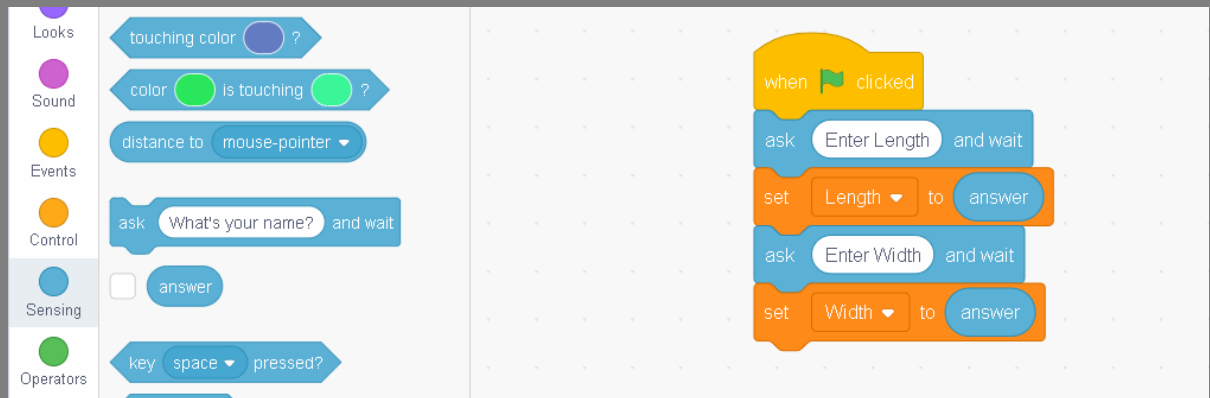
Saves the first number into the Length variable.



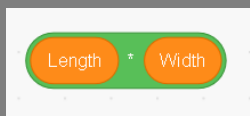
Prompts the user for the Width



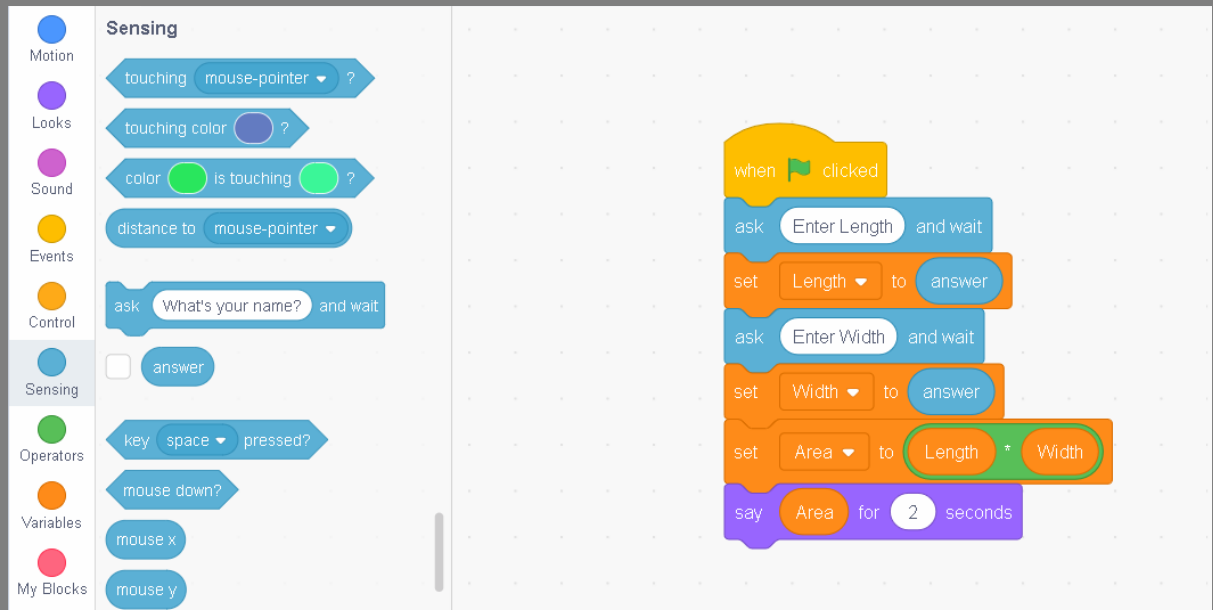
Saves the second number into the Width variable



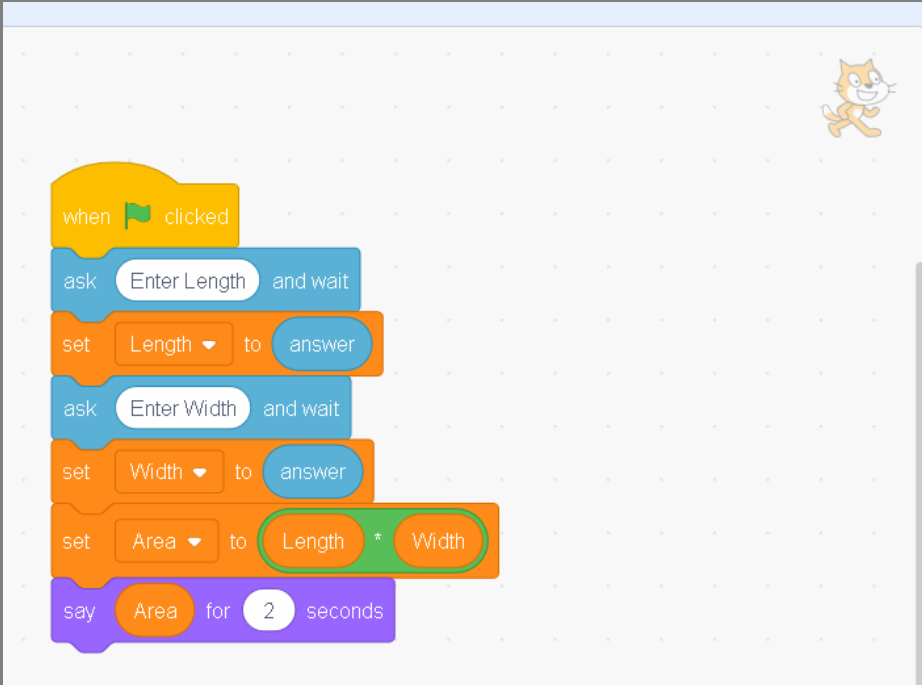
Constructing the maths block



Character will speak the final calculated number.



Output:



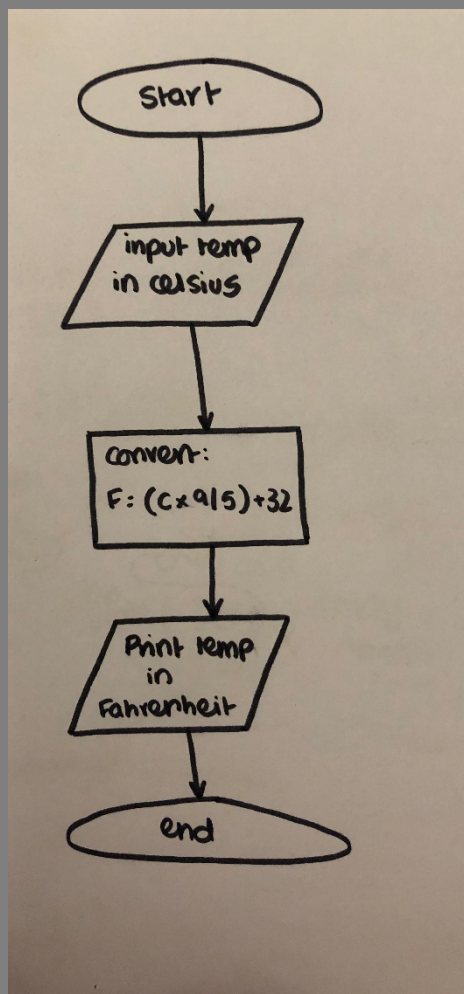
A Scratch script on a grid background. The script starts with a yellow 'when green flag clicked' block. It is followed by a blue 'ask' block with the text 'Enter Length' and 'and wait'. Then an orange 'set' block sets 'Length' to 'answer'. Next is another blue 'ask' block with 'Enter Width' and 'and wait'. Then an orange 'set' block sets 'Width' to 'answer'. This is followed by an orange 'set' block for 'Area' to the calculation 'Length * Width', where 'Length' and 'Width' are selected from dropdown menus. The final block is a purple 'say' block that says 'Area' for 2 seconds. On the right, a monitor shows 'Area' as 14, 'Length' as 7, and 'Width' as 2. The cat sprite is in the top right corner.

```
when green flag clicked
ask Enter Length and wait
set Length to answer
ask Enter Width and wait
set Width to answer
set Area to Length * Width
say Area for 2 seconds
```

Area 14
Length 7
Width 2

Sprite Sprite1

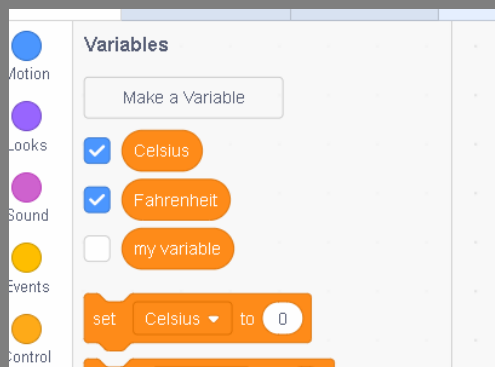
QUESTION 03



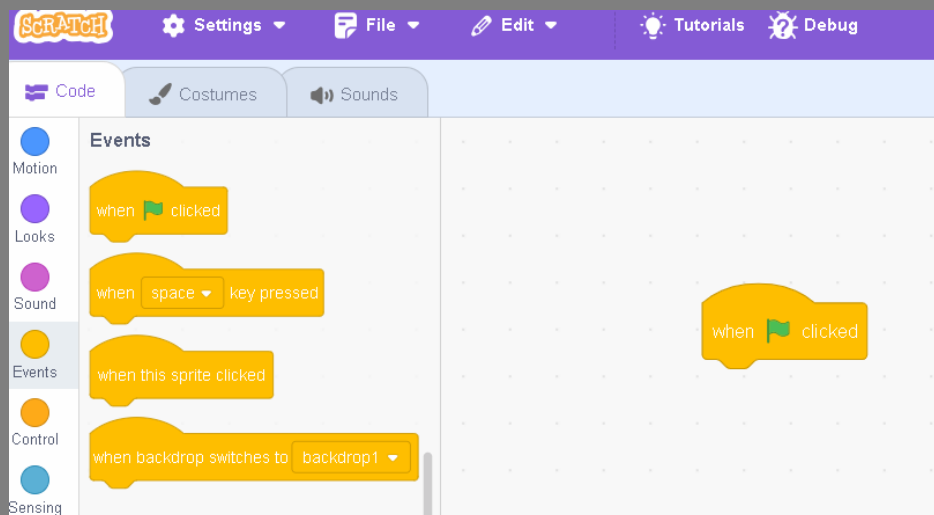
1. Start the program.
2. Declare two numeric variables: Celsius for input and Fahrenheit for output.
3. Ask the user to enter the temperature in Celsius.
4. Read the input and store it in Celsius.
5. Calculate the Fahrenheit value using the formula:

$$\text{Fahrenheit: } (\text{Celsius} \times 1.8) + 32$$
6. Store the result of the calculation in the Fahrenheit variable.
7. Display the value of Fahrenheit to the user.
8. End the program.

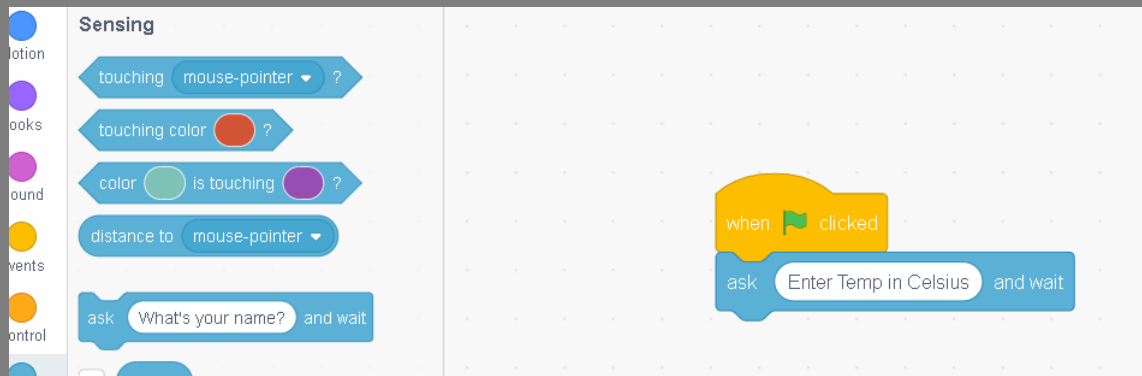
Creating variables that will store data



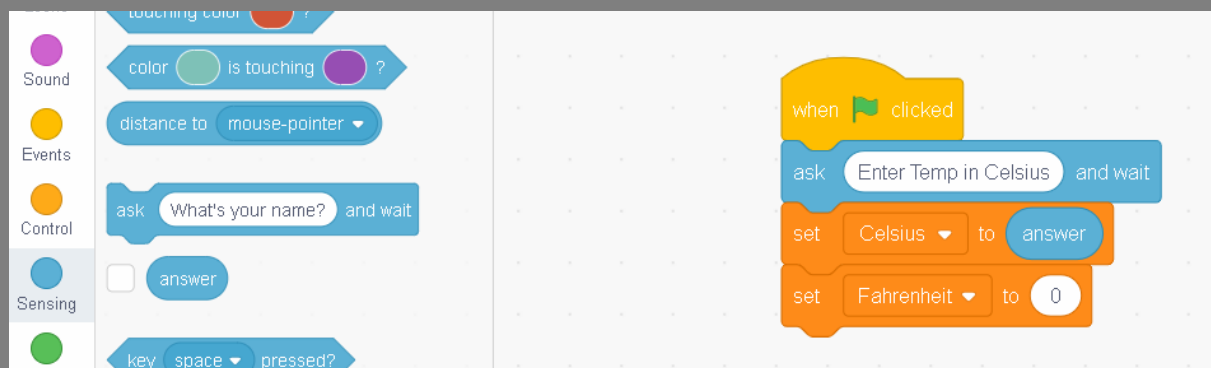
To start the program when you click the green flag



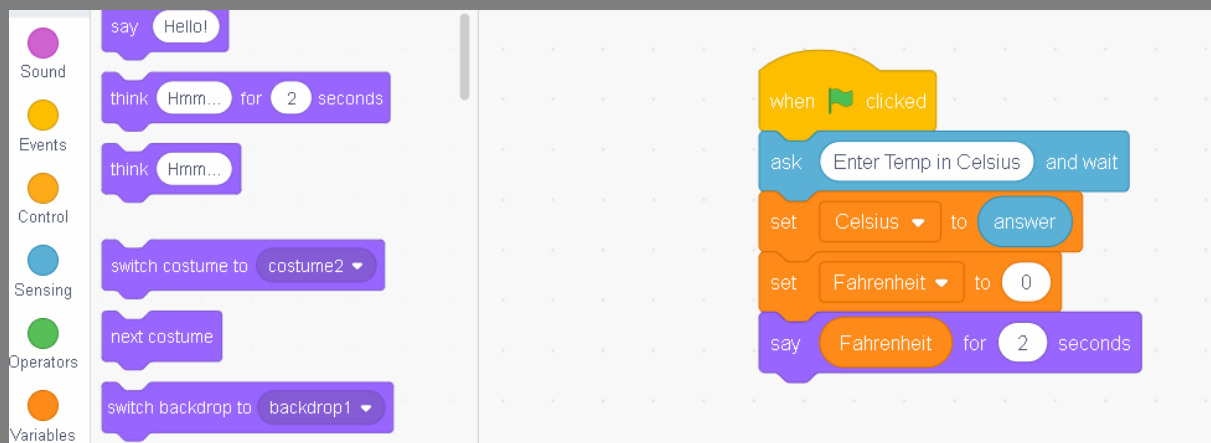
A text box will appear where user can type a number



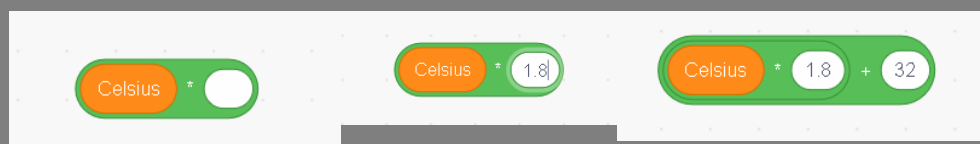
This takes the number you just typed and saves it inside the Celsius variable



Character will speak the final calculated number



Constructing the Math Formula using Operators blocks.



Output:

Scratch script and stage view.

Script Area:

- when green flag clicked
- ask Enter Temp in Celsius and wait
- set Celsius to answer
- set Fahrenheit to $\text{Celsius} * 1.8 + 32$
- say Fahrenheit for 2 seconds

Stage View:

- Celsius: 0
- Fahrenheit: 32

